# CSV content analysis

- number of URLs in the content
- number of dates in the content
- number of numeric values in the content
- the 100 most frequent words in the content
- a plot of the 10,000 most frequent words

```
from pathlib import Path
import re
from collections import Counter
import time

import duckdb
import pandas as pd
import matplotlib.pyplot as plt
```

## 1. Set the CSV path

Change the path below so it points to your CSV file.

```
CSV_PATH = Path(r"C:\Users\jespe\GDS eksamen\Data\NoteBooks\
news_cleaned_2018_02_13.csv")
CONTENT_COLUMN = "content"
ENCODING = "utf-8"
CHUNK_SIZE = 500000  # Process 500K rows at a time
```

## 2. Validate CSV header

This checks that the file is readable and that the target column exists, without loading the full dataset.

```
header_df = pd.read_csv(CSV_PATH, encoding=ENCODING, nrows=0)

if CONTENT_COLUMN not in header_df.columns:
    raise ValueError(
        f"Column '{CONTENT_COLUMN}' was not found. Available columns:
{list(header_df.columns)}"
    )

print(f"CSV is readable: {CSV_PATH}")
print(f"Detected columns: {list(header_df.columns)}")
print(f"Using chunk size: {CHUNK_SIZE:,} rows")
```

```
CSV is readable: C:\Users\jespe\GDS eksamen\Data\NoteBooks\
news_cleaned_2018_02_13.csv
```

```
Detected columns: ['Unnamed: 0', 'id', 'domain', 'type', 'url',
'content', 'scraped_at', 'inserted_at', 'updated_at', 'title',
'authors', 'keywords', 'meta_keywords', 'meta_description', 'tags',
'summary', 'source']
Using chunk size: 500,000 rows
```

## 3. Regex patterns

These patterns are practical defaults:

- **URL pattern**: matches `http://`, `https://`, and `www.`
- **Date pattern**: matches common formats such as:
  - `2024-03-11`
  - `11-03-2024`
  - `11/03/2024`
  - `March 11, 2024`
- **Numeric pattern**: matches integers, decimals, and signed values

```
URL_PATTERN = re.compile(
    r"\b(?:https?://|www\.)[^\s<>'\"()]+",
    flags=re.IGNORECASE
)

DATE_PATTERN = re.compile(
    r"\b(?:"
    r"\d{1,2}[/-]\d{1,2}[/-]\d{2,4}"
    r"|"
    r"\d{4}[/-]\d{1,2}[/-]\d{1,2}"
    r"|"
    r"(?:jan|feb|mar|apr|may|jun|jul|aug|sep|sept|oct|nov|dec)"
    r"[a-z]*\s+\d{1,2},?\s+\d{2,4}"
    r")\b",
    flags=re.IGNORECASE
)

NUMERIC_PATTERN = re.compile(
    r"(?<!\w)[+-]?(?:\d+(?:[.,]\d+)?)(?!\w)"
)

WORD_PATTERN = re.compile(r"\b[a-zA-ZÀ-ÿ']+\b")
```

## 4. Stream through CSV and aggregate counts

This processes the file chunk-by-chunk to avoid loading the full dataset into memory.

```
start_time = time.time()
print(f"Starting DuckDB processing at {time.strftime('%H:%M:%S')}\n")
```

```
# Create DuckDB connection with memory limit
con = duckdb.connect()
con.execute("SET memory_limit='8GB'")
con.execute("SET threads=8")

print("Step 1: Counting total rows...")
scan_start = time.time()

# Query directly without creating table to save memory
total_rows = con.execute(f"""
    SELECT COUNT(*)
    FROM read_csv_auto('{str(CSV_PATH)}',
                        sample_size=50000,
                        parallel=true,
                        strict_mode=false,
                        ignore_errors=true)
    WHERE content IS NOT NULL
""").fetchone()[0]

print(f"  ✓ Total rows: {total_rows:,} (scanned in {time.time() -
scan_start:.1f}s)\n")

# Step 2: Count URLs, dates, and numeric values - stream without
loading into table
print("Step 2: Counting URLs, dates, and numeric values...")
regex_start = time.time()

url_pattern = r'https?://[^\s<>''\"()]+|www\.[^\s<>''\"()]+'
date_pattern = r'\d{1,2}[/-]\d{1,2}[/-]\d{2,4}|\d{4}[/-]\d{1,2}[/-]\
d{1,2}'
numeric_pattern = r'\b\d+\.?\d*\b'

# Stream and count in one pass
result = con.execute(f"""
    SELECT
        SUM(LENGTH(regexp_extract_all(content, '{url_pattern}', 0)))
as total_urls,
        SUM(LENGTH(regexp_extract_all(content, '{date_pattern}', 0)))
as total_dates,
        SUM(LENGTH(regexp_extract_all(content, '{numeric_pattern}',
0))) as total_numeric
    FROM read_csv_auto('{str(CSV_PATH)}',
                        sample_size=50000,
                        parallel=true,
                        strict_mode=false,
                        ignore_errors=true)
    WHERE content IS NOT NULL
""").fetchone()

total_urls = result[0] if result[0] else 0
```

```
total_dates = result[1] if result[1] else 0
total_numeric = result[2] if result[2] else 0

print(f"  ✓ Done in {time.time() - regex_start:.1f}s")
print(f"    - URLs: {total_urls:,}")
print(f"    - Dates: {total_dates:,}")
print(f"    - Numeric: {total_numeric:,}\n")

# Step 3: Extract and count words in chunks
print("Step 3: Extracting and counting words...")
print("  (Processing in chunks to manage memory...)")
word_start = time.time()

word_counts = Counter()
chunk_size = 500000

for offset in range(0, total_rows, chunk_size):
    chunk_start_time = time.time()

    # Stream words from CSV chunk
    words_df = con.execute(f"""
        SELECT regexp_extract_all(LOWER(content), '[a-zA-ZÀ-ÿ'']+', 0)
as words
        FROM read_csv_auto('{str(CSV_PATH)}',
                           sample_size=50000,
                           parallel=true,
                           strict_mode=false,
                           ignore_errors=true)
        WHERE content IS NOT NULL
        LIMIT {chunk_size} OFFSET {offset}
    """).fetchdf()

    # Count words - handle arrays properly
    for word_list in words_df['words']:
        if word_list is not None and len(word_list) > 0:
            word_counts.update(word_list)

    processed = min(offset + chunk_size, total_rows)
    elapsed = time.time() - word_start
    remaining = (elapsed / processed) * (total_rows - processed) / 60
if processed > 0 else 0

    print(f"    - Processed {processed:,}/{total_rows:,} rows
({processed/total_rows*100:.1f}%) "
          f"in {elapsed:.1f}s - Est. remaining: {remaining:.1f} min")

    del words_df

print(f"  ✓ Word counting complete in {time.time() - word_start:.1f}s\
n")
```

```
# Step 4: Create preview
print("Step 4: Creating preview...")
preview_df = con.execute(f"""
    SELECT
        content,
        LENGTH(regexp_extract_all(content, '{url_pattern}', 0)) as
url_count,
        LENGTH(regexp_extract_all(content, '{date_pattern}', 0)) as
date_count,
        LENGTH(regexp_extract_all(content, '{numeric_pattern}', 0)) as
numeric_count
    FROM read_csv_auto('{str(CSV_PATH)}',
                       sample_size=50000,
                       parallel=true,
                       strict_mode=false,
                       ignore_errors=true)
    WHERE content IS NOT NULL
    LIMIT 5
""").fetchdf()

total_time = time.time() - start_time
print(f"\n{'='*60}")
print(f"PROCESSING COMPLETE!")
print(f"{'='*60}")
print(f"Total rows processed: {total_rows:,}")
print(f"Total time: {total_time/60:.1f} minutes ({total_time/3600:.2f}
hours)")
print(f"Average speed: {total_rows/total_time:.0f} rows/second")
print(f"{'='*60}\n")

# Close connection
con.close()

preview_df
```

```
Starting DuckDB processing at 15:02:29

Step 1: Counting total rows...

<>:29: SyntaxWarning: invalid escape sequence '\.'
<>:29: SyntaxWarning: invalid escape sequence '\.'
C:\Users\jespe\AppData\Local\Temp\ipykernel_9108\1108022216.py:29:
SyntaxWarning: invalid escape sequence '\.'
  url_pattern = r'https?://[^\s<>''\"()]+|www\.[^\s<>''\"()]+'

  ✓ Total rows: 8,529,477 (scanned in 26.6s)

Step 2: Counting URLs, dates, and numeric values...
  ✓ Done in 65.9s
    - URLs: 2,285,422
```

```
    - Dates: 696,215
    - Numeric: 79,323,243

Step 3: Extracting and counting words...
  (Processing in chunks to manage memory...)
    - Processed 500,000/8,529,477 rows (5.9%) in 196.5s - Est.
remaining: 52.6 min
    - Processed 1,000,000/8,529,477 rows (11.7%) in 609.0s - Est.
remaining: 76.4 min
    - Processed 1,500,000/8,529,477 rows (17.6%) in 972.4s - Est.
remaining: 75.9 min
    - Processed 2,000,000/8,529,477 rows (23.4%) in 1445.2s - Est.
remaining: 78.6 min
    - Processed 2,500,000/8,529,477 rows (29.3%) in 2180.1s - Est.
remaining: 87.6 min
    - Processed 3,000,000/8,529,477 rows (35.2%) in 2964.2s - Est.
remaining: 91.1 min
    - Processed 3,500,000/8,529,477 rows (41.0%) in 3677.9s - Est.
remaining: 88.1 min
    - Processed 4,000,000/8,529,477 rows (46.9%) in 4364.7s - Est.
remaining: 82.4 min
    - Processed 4,500,000/8,529,477 rows (52.8%) in 5031.3s - Est.
remaining: 75.1 min
    - Processed 5,000,000/8,529,477 rows (58.6%) in 5690.1s - Est.
remaining: 66.9 min
    - Processed 5,500,000/8,529,477 rows (64.5%) in 6493.2s - Est.
remaining: 59.6 min
    - Processed 6,000,000/8,529,477 rows (70.3%) in 7259.8s - Est.
remaining: 51.0 min
    - Processed 6,500,000/8,529,477 rows (76.2%) in 8199.5s - Est.
remaining: 42.7 min
    - Processed 7,000,000/8,529,477 rows (82.1%) in 9076.8s - Est.
remaining: 33.1 min
    - Processed 7,500,000/8,529,477 rows (87.9%) in 10126.2s - Est.
remaining: 23.2 min
    - Processed 8,000,000/8,529,477 rows (93.8%) in 11421.2s - Est.
remaining: 12.6 min
    - Processed 8,500,000/8,529,477 rows (99.7%) in 12555.5s - Est.
remaining: 0.7 min
    - Processed 8,529,477/8,529,477 rows (100.0%) in 13530.2s - Est.
remaining: 0.0 min
  ✓ Word counting complete in 13530.5s

Step 4: Creating preview...

============================================================
PROCESSING COMPLETE!
============================================================
Total rows processed: 8,529,477
Total time: 227.1 minutes (3.78 hours)
```

```
Average speed: 626 rows/second
==============================================================


                                                  content  url_count
date_count  \
0  Life is an illusion, at least on a quantum lev...          0
0
1  Unfortunately, he hasn’t yet attacked her for ...          0
0
2  The Los Angeles Police Department has been den...          0
0
3  The White House has decided to quietly withdra...          0
0
4  “The time has come to cut off the tongues of t...          0
0

   numeric_count
0              1
1              0
2              1
3              0
4              0
```

## 5. Show total counts for the whole dataset

```
print(f"Total URLs found: {total_urls:,}")
print(f"Total dates found: {total_dates:,}")
print(f"Total numeric values found: {total_numeric:,}")

Total URLs found: 2,285,422
Total dates found: 696,215
Total numeric values found: 79,323,243
```

## 6. Find the 100 most frequent words in the content

Word frequencies were aggregated during chunked processing above.

```
top_100_words = word_counts.most_common(100)
top_100_df = pd.DataFrame(top_100_words, columns=["word", "count"])

top_100_df

     word      count
0     the  224561604
1      to  108430614
2      of  107081534
3     and   98657259
4       a   86002693
```

```
..    ...        ...
95  trump    4084417
96    any    3934880
97  obama    3905724
98  world    3883279
99    may    3835844

[100 rows x 2 columns]
```

```python
pd.set_option('display.max_rows', 100)
print(top_100_df)
```

```
          word      count
0          the  224561604
1           to  108430614
2           of  107081534
3          and   98657259
4            a   86002693
5           in   74778922
6         that   48035024
7           is   43466104
8          for   37619987
9           on   30228375
10          it   28789270
11           s   28507299
12        with   24277251
13          as   24028366
14         was   20956755
15         are   20942371
16          by   20104451
17           i   19648746
18        this   19549899
19          at   18897455
20          be   18633131
21         not   18103194
22          he   18081808
23         you   18003314
24        have   17998641
25        from   17498211
26          an   14795685
27         has   14697573
28        they   14387808
29         but   14137143
30         his   13963358
31          we   13726089
32          or   13108743
33        said   12222982
34         who   11538303
35        will   11402474
36       their   10862361
```

| | | |
|---|---|---|
| 37 | all | 10224760 |
| 38 | more | 9955273 |
| 39 | one | 9562230 |
| 40 | about | 9505556 |
| 41 | new | 9160420 |
| 42 | which | 8803867 |
| 43 | if | 8494556 |
| 44 | up | 8413661 |
| 45 | would | 8295647 |
| 46 | can | 8208065 |
| 47 | what | 7971621 |
| 48 | been | 7936756 |
| 49 | had | 7843681 |
| 50 | were | 7837737 |
| 51 | people | 7669585 |
| 52 | there | 7575051 |
| 53 | so | 7506797 |
| 54 | when | 7337638 |
| 55 | out | 7262516 |
| 56 | our | 7076967 |
| 57 | t | 7021396 |
| 58 | no | 6765159 |
| 59 | its | 6504827 |
| 60 | like | 6193734 |
| 61 | your | 5987035 |
| 62 | also | 5943806 |
| 63 | her | 5862724 |
| 64 | than | 5845405 |
| 65 | us | 5797920 |
| 66 | do | 5711620 |
| 67 | she | 5652878 |
| 68 | after | 5620598 |
| 69 | other | 5611575 |
| 70 | time | 5533176 |
| 71 | some | 5494934 |
| 72 | just | 5350578 |
| 73 | over | 5186082 |
| 74 | now | 5179074 |
| 75 | them | 5138985 |
| 76 | into | 5106614 |
| 77 | my | 5061546 |
| 78 | year | 4939018 |
| 79 | only | 4930798 |
| 80 | mr | 4904420 |
| 81 | two | 4622627 |
| 82 | even | 4608656 |
| 83 | first | 4465488 |
| 84 | state | 4425575 |
| 85 | how | 4396615 |

```
86        news    4352368
87   president    4328592
88        most    4302111
89     because    4293955
90       these    4230988
91       could    4201007
92        many    4179672
93       years    4133368
94  government    4095874
95       trump    4084417
96         any    3934880
97       obama    3905724
98       world    3883279
99         may    3835844
```

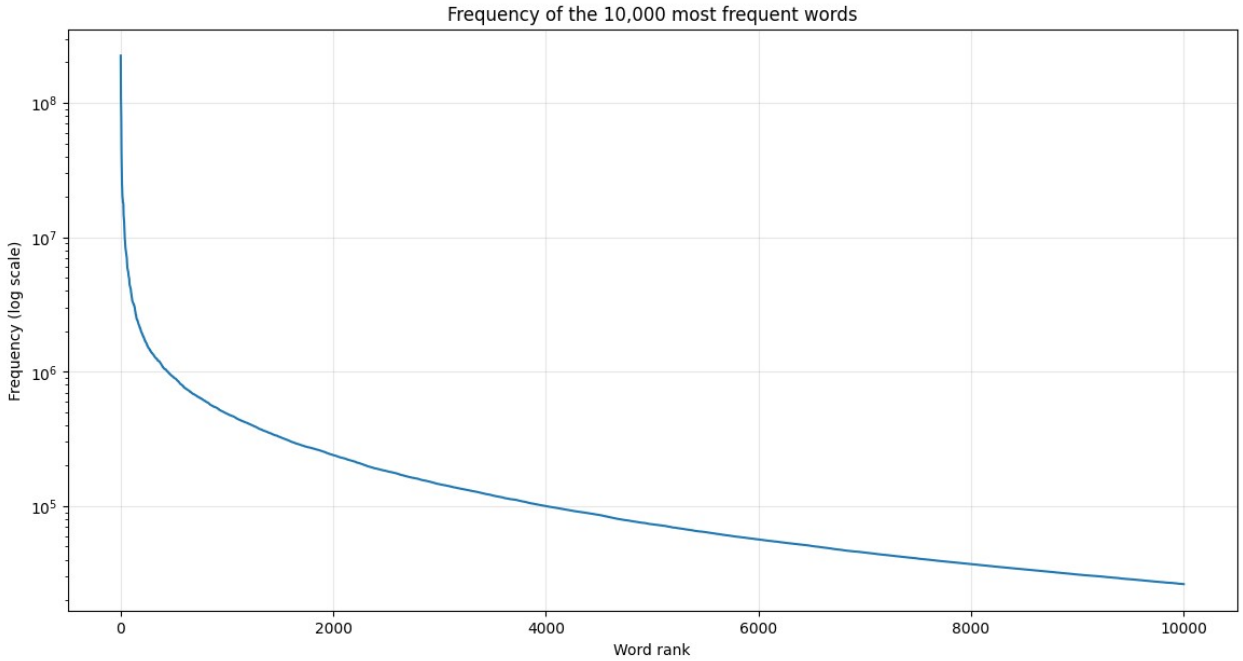
## 7. Plot the frequency of the 10,000 most frequent words

Because frequencies usually drop very quickly, the y-axis is shown on a logarithmic scale.

```
top_10000 = word_counts.most_common(10000)

freq_values = [count for _, count in top_10000]
ranks = range(1, len(freq_values) + 1)

plt.figure(figsize=(14, 7))
plt.plot(ranks, freq_values)
plt.yscale("log")
plt.xlabel("Word rank")
plt.ylabel("Frequency (log scale)")
plt.title("Frequency of the 10,000 most frequent words")
plt.grid(True, alpha=0.3)
plt.show()
```


Frequency of the 10,000 most frequent words
Frequency (log scale)
$10^8$
$10^7$
$10^6$
$10^5$
0
2000
4000
6000
8000
10000
Word rank